Lecture 13 — Oscillations and Finite Differences in Time

18.085 Computational Science and Engineering I • MW 9:30–11, 6-120 • Wednesday, October 28, 2026

After this lecture you should be able to

- Turn the static equilibrium $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{0}$ into Newton's law $\boldsymbol{M}\ddot{\boldsymbol{u}} + \boldsymbol{K}\boldsymbol{u} = \boldsymbol{0}$, arguing the sign of the internal spring force rather than guessing it.
- Diagonalise $\boldsymbol{K}$ to decouple the motion into normal modes oscillating at $\omega_j = \sqrt{\lambda_j}$.
- Reduce a second-order equation to a first-order dynamical system and read its trajectory in phase space.
- Build the growth matrix $\boldsymbol{G}$ for forward Euler and read from its eigenvalues $|\lambda(\boldsymbol{G})|$ why the computed oscillation grows.

1 From equilibrium to motion

In the last lecture we learned how to solve for the *equilibrium state* of a spring–mass system. Let us now look at *time dynamics*: the chain is not sitting still, it is oscillating, and we want to know how.

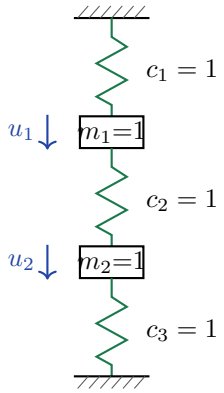
For simplicity, assume there are no external forces at all. The static equilibrium equation is then

$$\boldsymbol{K}\boldsymbol{u} = \boldsymbol{0}.$$

The dynamics come from Newton's second law,

$$\boldsymbol{f} = \boldsymbol{M}\boldsymbol{a},$$

and the acceleration is just the second time derivative of the displacement, $\boldsymbol{a} = \ddot{\boldsymbol{u}}$. The only real question is what $\boldsymbol{f}$ should be on the left-hand side. In the absence of external forces we have nothing but the *internal* forces produced by the springs.



Getting the sign right

Previously, when we *did* have external forces, the equation was

$$\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}_{\text{ext}}, \qquad f_{\text{ext},i} \text{ positive pointing } \textit{downwards}.$$

That tells us how to read $\boldsymbol{Ku}$. The spring force on mass i is $(\boldsymbol{Ku})_i$, and it points *upwards* when positive — that is exactly how it balances the downwards positive $f_{\text{ext},i}$.

On the other hand u_i is positive for a *downwards* displacement, so $a_i = \ddot{u}_i$ is positive for a *downwards* acceleration. The two sides of Newton's law therefore have opposite orientations, and we must put a minus sign in:

$$-\boldsymbol{Ku} = \boldsymbol{M\ddot{u}} \qquad \Longrightarrow \qquad \boxed{\boldsymbol{M\ddot{u} + Ku = 0}}.$$

The minus sign is physics, not bookkeeping

If you write $\boldsymbol{M\ddot{u} = Ku}$ you get exponential growth and decay instead of oscillation — a rod that flies apart rather than one that rings. The sign comes from $\boldsymbol{K}$ measuring the *restoring* force: displace a mass downwards and the springs pull it back up. Always trace it through the sign convention on $\boldsymbol{f}_{\text{ext}}$ rather than trusting memory.

Newton's law, one mass at a time

The same equation comes out if we write Newton's second law for each mass separately.

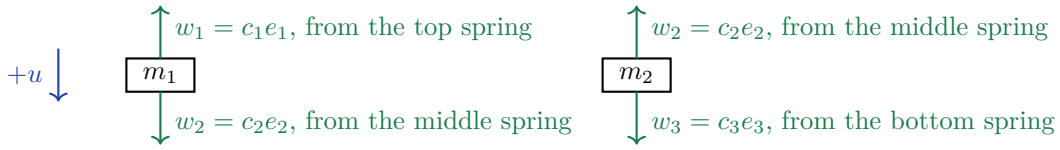
Example 1.1 (Two masses on three springs)**. Given.** The chain of the figure with any masses m_1, m_2 and spring constants c_1, c_2, c_3, no external forces, and u_1, u_2 positive downwards.

Find. What is Newton's second law for each mass? What are $\boldsymbol{M}$ and $\boldsymbol{K}$?

Answer. Spring j is stretched by its elongation e_j, and it pulls both of its ends towards its middle with the tension $w_j = c_j e_j$. The ceiling and the floor do not move, so

$$e_1 = u_1, \qquad e_2 = u_2 - u_1, \qquad e_3 = -u_2;$$

the bottom spring is squeezed ($e_3 < 0$) when mass 2 moves down. The forces on each mass, drawn for $e_j > 0$:



Counting downwards as positive, Newton's second law for each mass reads

$$m_1\ddot{u}_1 = -w_1 + w_2 = -c_1u_1 + c_2(u_2 - u_1),$$
$$m_2\ddot{u}_2 = -w_2 + w_3 = -c_2(u_2 - u_1) - c_3u_2.$$

Move the spring forces to the left and collect u_1 and u_2:

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{bmatrix} + \begin{bmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 + c_3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This is $\boldsymbol{M\ddot{u} + Ku = 0}$ with $\boldsymbol{M} = \operatorname{diag}(m_1, m_2)$ and the stiffness matrix $\boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{CA}$ of the last lecture, built from

$$\boldsymbol{e} = \boldsymbol{Au}, \quad \boldsymbol{w} = \boldsymbol{Ce}, \qquad \boldsymbol{A} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix}, \quad \boldsymbol{C} = \operatorname{diag}(c_1, c_2, c_3).$$

In this language the two Newton equations read $\boldsymbol{M\ddot{u}} = -\boldsymbol{A}^{\mathsf{T}}\boldsymbol{w} = -\boldsymbol{A}^{\mathsf{T}}\boldsymbol{CAu}$. The net upward spring force on mass i is $(\boldsymbol{Ku})_i$, as the sign argument said: the minus sign appears by itself, because every spring pulls against its own stretch.

If gravity acts, measure $\boldsymbol{u}$ from the equilibrium position $\boldsymbol{u}_{\text{eq}}$. There the weights are balanced, $\boldsymbol{K}\boldsymbol{u}_{\text{eq}} = \boldsymbol{f}$, so $\boldsymbol{u} - \boldsymbol{u}_{\text{eq}}$ obeys the same equation.

For the chain in the figure, with every m_i and c_j equal to 1,

$$\boldsymbol{K} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \qquad \boldsymbol{M} = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix} = \boldsymbol{I}.$$

Having the mass matrix be the identity simplifies the discussion without loss of generality, and leaves us with

$$\ddot{\boldsymbol{u}} + \boldsymbol{K}\boldsymbol{u} = \boldsymbol{0}.$$

How do we solve this?

2 Normal modes: choose a basis adapted to $\boldsymbol{K}$

One way to analyse the system: pick a nice basis with respect to the matrix $\boldsymbol{K}$. The obvious candidate is the *eigenbasis* of $\boldsymbol{K}$. Let the eigendecomposition be

$$\boldsymbol{K} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^{\mathsf{T}},$$

which we may do because $\boldsymbol{K}$ is symmetric — so $\boldsymbol{Q}$ is orthogonal and $\boldsymbol{\Lambda}$ is real and diagonal. Substituting,

$$\ddot{\boldsymbol{u}} + \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{u} = \boldsymbol{0} \qquad \Longrightarrow \qquad \boldsymbol{Q}^{\mathsf{T}}\ddot{\boldsymbol{u}} + \boldsymbol{\Lambda}(\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{u}) = \boldsymbol{0},$$

after multiplying through by $\boldsymbol{Q}^{\mathsf{T}}$ on the left. Now redefine

$$\boldsymbol{x} \equiv \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{u},$$

so $\boldsymbol{x}$ is the coefficient vector of $\boldsymbol{u}$ in the eigenbasis $\boldsymbol{Q}$ of the stiffness matrix $\boldsymbol{K}$. The equation becomes

$$\ddot{\boldsymbol{x}} + \boldsymbol{\Lambda}\boldsymbol{x} = \boldsymbol{0}, \qquad \text{i.e.} \qquad \frac{d^2}{dt^2}\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \boldsymbol{0},$$

and, because $\boldsymbol{\Lambda}$ is diagonal, **the system is decoupled**:

$$\begin{cases} \dfrac{d^2}{dt^2}x_1 + \lambda_1 x_1 = 0 \\ \qquad\qquad\qquad\quad \vdots \\ \dfrac{d^2}{dt^2}x_n + \lambda_n x_n = 0. \end{cases}$$

Diagonalising is decoupling

n masses joined by springs are n coupled oscillators; in the eigenbasis of $\boldsymbol{K}$ they are n *independent* oscillators that never talk to each other. Every hard question about the coupled chain becomes n easy questions about a single mass on a single spring. This is the same superposition move as in Lecture 1, wearing a mechanical costume.

Solving the decoupled equations

Each line is the scalar equation $\ddot{x} + \lambda x = 0$, whose solutions are sines and cosines. Defining

$$\omega_1 = \sqrt{\lambda_1}, \quad \ldots, \quad \omega_n = \sqrt{\lambda_n},$$

the solutions to the decoupled equations are

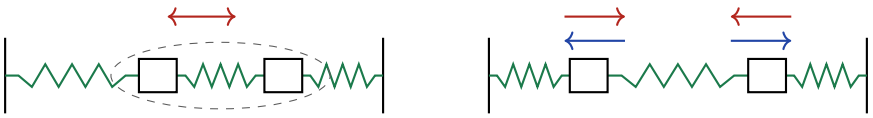
$$x_1 = a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t), \quad \ldots, \quad x_n = a_n \cos(\omega_n t) + b_n \sin(\omega_n t),$$

where the coefficients $\{a_j\}$ and $\{b_j\}$ are determined by the initial conditions.

To get $\boldsymbol{u}$ back, we basis-transform in the other direction: $\boldsymbol{x} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{u} \Rightarrow \boldsymbol{u} = \boldsymbol{Q}\boldsymbol{x}$, so

$$\boldsymbol{u}(t) \;=\; \sum_{j=1}^{n} \boldsymbol{q}_j \big[\, a_j \cos(\omega_j t) + b_j \sin(\omega_j t) \,\big].$$

Read the two halves of that sum separately. The eigenvector $\boldsymbol{q}_j$ carries the **spatial dependence** — the *shape* the chain takes. The bracket carries the **time dependence**, an oscillation at the frequency ω_j of *normal mode* j. The general motion is a superposition of normal modes.



normal mode 1 normal mode 2

Why engineers care about ω_j

Each $\omega_j = \sqrt{\lambda_j}$ is a resonant frequency of the structure, and $\boldsymbol{q}_j$ is the shape it resonates in. The eigenvalue problem we solved abstractly in Lectures 6–8 is, in this setting, the question of what a building or a bridge does when the ground shakes at a particular frequency.

3 Finite differences in time

The mass–spring system is also a good toy problem for touching upon *finite difference methods for time-based differential equations*, also known as **dynamical systems**.

For simplicity, let us shrink all the way down to one mass and one spring. The system is

$$\ddot{u} + u = 0,$$

which is a *second-order* system.

First insight: lower the order by adding variables

We can lower the order of the system by introducing new variables. Define

$$v \equiv \dot{u}, \qquad \text{i.e.} \qquad \frac{du}{dt} = v,$$

and then $\ddot{u} + u = 0$ says exactly

$$\frac{dv}{dt} = -u.$$

Taken together we now have *only first-order* time derivatives:

$$\frac{d}{dt}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix}.$$

Interpretation

v is the velocity of the mass on the spring. Instead of a single second-order equation in the displacement u alone, we are studying an equivalent *system* of first-order equations in u and v.

In general, a system of first-order differential equations with respect to time is called a **dynamical system**. In principle any kind of higher-order time dynamics can be reduced to a dynamical system with only first derivatives in time, through the introduction of new variables such as v.

One canonical form

Order is not a fundamental property of a time-dependent problem — it is a choice of variables. Reduce everything to $\dot{\boldsymbol{y}} = \boldsymbol{A}\boldsymbol{y}$ and you only ever need *one* family of time-stepping methods, applied to a vector.

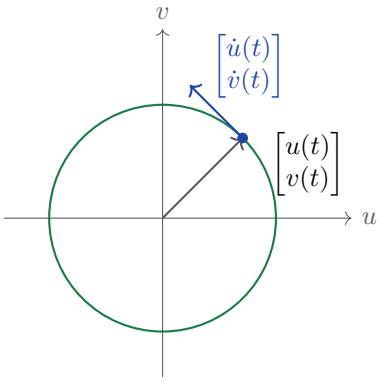
Phase space

To visualise the time evolution of dynamical systems we often plot the trajectories of the dependent variables in a **phase space diagram**. "Phase space" is simply a technical term for the possible values the dependent variables can take on.

For the mass–spring system, $\dot{u} = v$ and $\dot{v} = -u$. Note that

$$\begin{bmatrix} \dot{u} & \dot{v} \end{bmatrix} \cdot \begin{bmatrix} u \\ v \end{bmatrix} = \dot{u}\,u + \dot{v}\,v = (v)\,u + (-u)\,v = uv - uv = 0.$$

So the time derivative of the vector $\begin{bmatrix} u \\ v \end{bmatrix}$ is *always orthogonal to it.* A vector whose rate of change is perpendicular to itself never changes length. Therefore the phase-space trajectory $\begin{bmatrix} u(t) \\ v(t) \end{bmatrix}$ is a **circle**.



The circle is conservation of energy

The radius $\sqrt{u^2 + v^2}$ is fixed by the energy in the system, which the true dynamics conserve. Everything interesting about the numerical methods below is a statement about what they do to that radius.

4 Discretising time

Suppose we now discretise time. Put

$$t_n = n\,\Delta t, \qquad U_n \equiv u(t_n), \qquad V_n \equiv v(t_n),$$

and replace each time derivative by a forward difference,

$$\frac{du}{dt}\Big|_{t_n} \simeq \frac{U_{n+1} - U_n}{\Delta t}, \qquad \frac{dv}{dt}\Big|_{t_n} \simeq \frac{V_{n+1} - V_n}{\Delta t}.$$

So the two equations of the dynamical system become

$$\frac{U_{n+1} - U_n}{\Delta t} = v, \qquad \frac{V_{n+1} - V_n}{\Delta t} = -u.$$

The whole question in one line

What grid points should we use on the right-hand side? The left-hand sides are settled; the right-hand sides still say u and v with no subscript. Different choices of which time level to evaluate them at lead to different methods — and to completely different long-time behaviour.

Forward Euler

Take everything on the right at the *old* time level:

$$\text{RHS} = \begin{bmatrix} V_n \\ -U_n \end{bmatrix}.$$

We obtain

$$\begin{cases} \dfrac{U_{n+1} - U_n}{\Delta t} = V_n \\ \dfrac{V_{n+1} - V_n}{\Delta t} = -U_n \end{cases} \quad \Longrightarrow \quad \begin{cases} U_{n+1} = U_n + \Delta t\, V_n \\ V_{n+1} = V_n - \Delta t\, U_n \end{cases}$$

which is *explicit*: the new values are written down directly from the old ones. In matrix form,

$$\begin{bmatrix} U_{n+1} \\ V_{n+1} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \Delta t \\ -\Delta t & 1 \end{bmatrix}}_{\boldsymbol{G}_F} \begin{bmatrix} U_n \\ V_n \end{bmatrix},$$

and $\boldsymbol{G}_F$ is called the **growth matrix**. The end result of n steps is

$$\begin{bmatrix} U_n \\ V_n \end{bmatrix} = \boldsymbol{G}_F^{\,n} \begin{bmatrix} U_0 \\ V_0 \end{bmatrix}, \qquad \text{with } \begin{bmatrix} U_0 \\ V_0 \end{bmatrix} \text{ the initial conditions.}$$

Qualitatively, what happens to $\begin{bmatrix} U_n \\ V_n \end{bmatrix}$ as n gets large depends on the *eigenvalues* of $\boldsymbol{G}_F$. Writing $h = \Delta t$ for brevity, $\boldsymbol{G}_F$ has eigenvalues

$$\lambda = 1 + ih, \qquad \lambda = 1 - ih, \qquad \text{so} \qquad |\lambda| = \sqrt{1 + h^2} > 1,$$

and we will see a *growth* in the size of the vector $\begin{bmatrix} U_n \\ V_n \end{bmatrix}$.

Where this is going

Forward Euler puts energy into the oscillator at every step. Next time we pick up from here with the other choices of grid points on the right-hand side: backward Euler, leapfrog and the trapezoidal method. Then we add the fourth-order Runge–Kutta method and compare all of them on the same problem.

Reading

Strang, *Computational Science and Engineering*, § 2.2 (Oscillation by Newton's Law).